1 Introduction

History

The development of the first transistor in 1948 was was the start of the micro-electronic evolution. The early commercial device was a germanium junction transistor. Although germanium has higher electron and hole mobility than silicon, germanium transistors suffer from several inherent disadvantages, which are as follows.

- Poor temperature characteristic than silicon results in high leakage of current at elevated temperatures.
- There is inability to protect the junction between p and n regions from oxygen or water exposure.
- There is a requirement of very effective hermetically sealed packages.
- The gradual development of silicon technology opened the way to integrating complete circuits on silicon.

Evolution

During the first two decades, chip complexity increased from small scale integration (SSI) through medium scale integration (MSI) and large scale integration (LSI) to very large scale integration (VLSI). This is illustrated in Fig. 1.1.

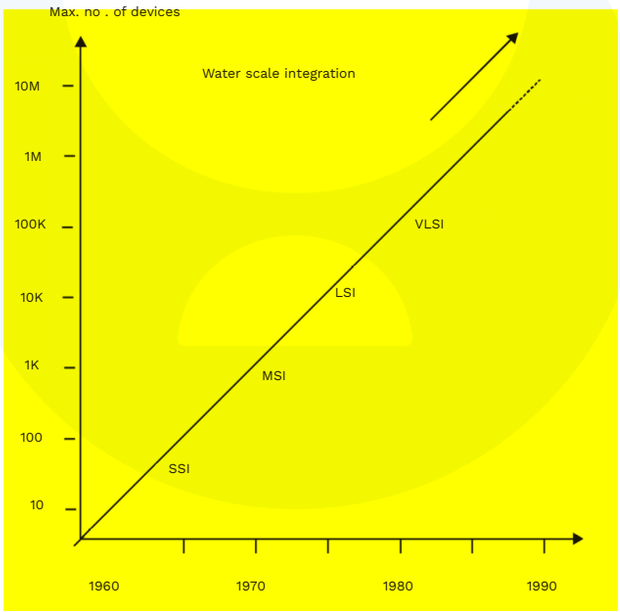


Fig. 1.1 The Classic Moore's Law Graph Showing an Increase In Maximum Possible Capability per Single LC Chip since 1960. The SSI is Usually Taken as Tens of Transistors, MSI Hundreds of Transistors, LSI Thousands of Transistors, and VLSI Tens of Thousands of Transistors

Since the silicon area required for transistors is less compared to resistors and capacitors, the impact of this evolution on electronic circuit design was to encourage the use of active rather than passive devices. The two technologies developed over this period were bipolar and metal-oxide silicon (MOS).

The significant features of each technology may be summarised as follows.

Bipolar

- It is faster than MOS, but it usually takes more power.
- There is higher transconductance (gm) than MOS. Thus, it has a better signal amplification performance.
- There are fewer fabrication steps than in MOS/CMOS fabrication techniques.

MOS

- There is higher impedance levels than in bipolar, thus, reducing power dissipation.
- Device performance (which depends on the and L of transistors) can be tailored during the chip layout stage, a facility absent in bipolar design.

Digital complementary metal oxide semiconductors (CMOS) ICs are the driving force behind VLSI for high performance computing and other scientific and engineering applications due to their salient features, such as low power, reliable performance, circuit techniques for high speed, and ongoing improvements in processing technology.

Bipolar and gallium arsenide (GaAs) are inefficient for VLSI due to processing difficulties and high power consumption, although they are used for special applications.

VLSI Design Methodologies

There are two design methodologies for VLSI circuits.

- Full custom design style
- Semi-custom design style

In the entire custom design style, the geometry and placement of every transistor can be optimised individually, whereas in the semi-custom design style (Standard Cell Based Design or field-programmable gate array), the used components are already optimised. There is a clear difference between the two styles as shown in the following table.

S. No.	Full Custom	Semi-custom
1.	It is a time-consuming or slow.	The design time is comparatively shorter.
2.	There are more opportunities for circuit performance improvement.	It has a comparatively lower performance than full custom.
3.	It has high reliability.	It has low reliability.
4.	It is expensive.	It is cost-effective and less time-consuming.

Table 1.1

The choice of design style depends on the performance requirements, technology used (W/L ratio of transistor), expected lifetime of the product, and project cost. Some considerations to be kept in mind while designing include speed, power consumption, cost, efficiency, reliability of the product, etc.

VLSI Design Flow

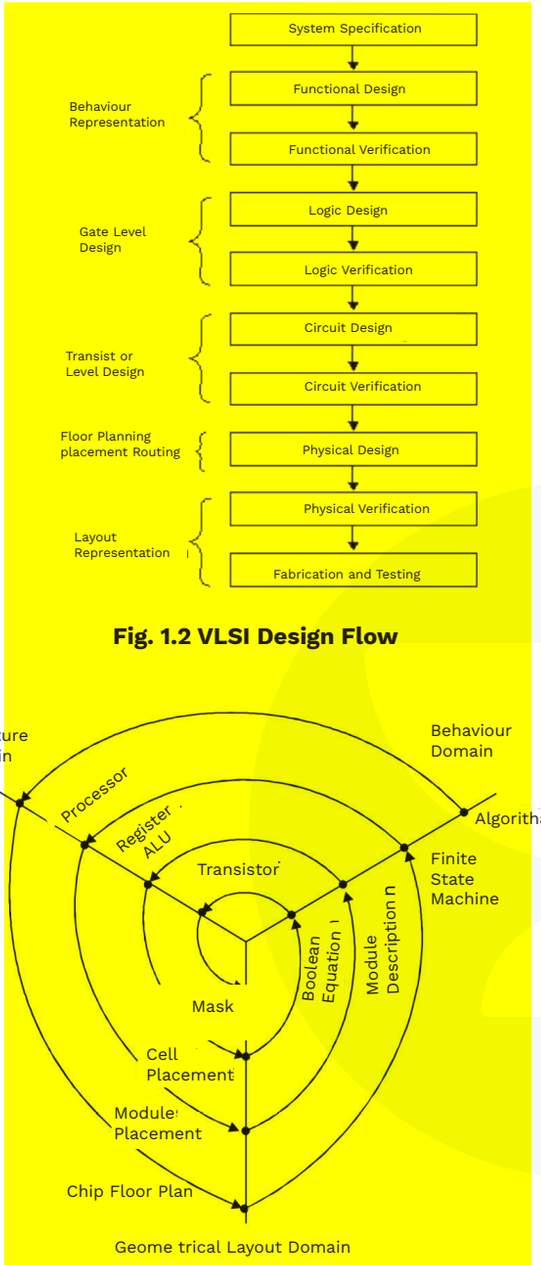


Fig. 1.2 VLSI Design Flow

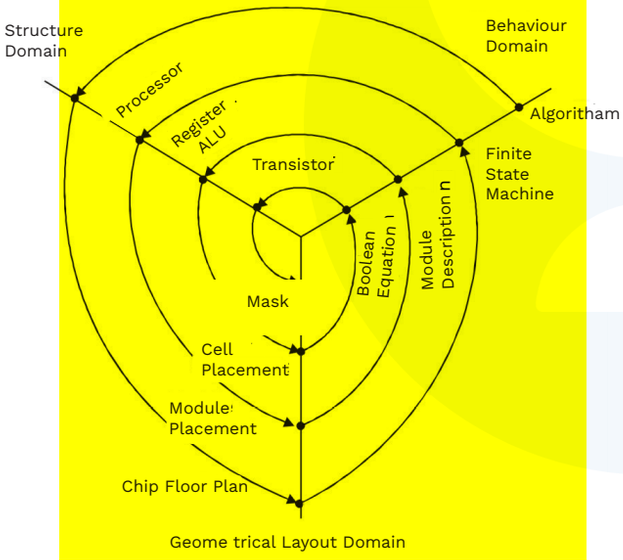


Fig. 1.3 Y-Chart (Gajski chart)

The Y-shaped chart (Y-chart) represents the three domains of design flow.

The design flow starts with an algorithm that describes the required chip's behaviour. According to the algorithm, the corresponding processor's architecture is defined, which is then mapped onto the chip surface by floor planning. The next design evolution describes the Finite State Machine (FSM), which is structurally implemented using registers and Arithmetic Logic Units (ALUs). These modules are then geometrically placed onto the chip surface using placement and routing methods. The third evolution starts with a behavioural module description. Each module is then implemented with leaf cells (logic gates), which are then placed and interconnected by using a cell placement and routing program.

The latest evolution provides a detailed Boolean description of leaf cells followed by transistor level implementation of the leaf cells and mask generation.

In the standard cell based design style, leaf cells are pre-designed (at the transistor level) and stored in a library for logic implementation, and thus, eliminates the need for transistor-level design.

Fig. 1.2 illustrates a simplified view of the design flow. Both top-down and bottom-up approaches are combined for a successful design, for example, if a chip designer defines an architecture that may exceed the available chip area, then some functions have to be removed and the design process must be repeated. Therefore, it is essential to feed forward low level information a higher level (bottom up) as early as possible. Some classical techniques for reducing the complexity of IC design are hierarchy, regularity, modularity, and locality.

Concept of Hierarchy, Regularity, Modularity, and Locality

Design hierarchy: it refers to divide and conquer technique. Dividing a module into modules and sub-modules until the complexity of the smaller parts becomes manageable.

Regularity: it refers to the hierarchical decomposition of a large system that must result in not only simple but also similar blocks as much as possible.

- There is regularity at all levels of abstraction.
- At the transistor level, uniformly sized transistors simplify the design and at the logic level, identical gate structures can be used.

Modularity: it means that the various functional blocks that constitute the larger system must have well-defined functions and interfaces.

- It allows the use of generic modules in various designs.
- All blocks can be easily combined towards the end to form the large system.

Locality: it infers avoiding long distance connections as much as possible (to avoid long interconnect delays).
Connection mostly between neighbouring modules.

Practice Exercise

Level I

1. CMOS stands for:
(A) Complex Metal Oxide Semiconductor
(B) Complementary Metal Oxide Semiconductor
(C) Common Metal On Semiconductor
(D) Complex Metal On Semiconductor

2. Choose the correct answer.
I. BJT is chosen over MOS in VLSI circuits.
II. MOS has better speed than BJT.
III. BJT is faster than MOS but consumes more power.
IV. MOS based circuits consume less power.
(A) I and II only
(B) IV only
(C) Both III and IV
(D) Both I and IV

3. Which of the following statements is true regarding VLSI design flow?
(A) Floorplanning and routing are the parts of functional design.
(B) Testing and verification is required to be done after each level of design flow.
(C) Logic design means designing using logic gates.
(D) Physical design considers the placement of cells only.

4. Which step is required after each level of design flow?
(A) Verification
(B) Testing
(C) Analysis
(D) Monitoring

5. Choose the correct order of the following steps.
I. Architecture design
II. Market requirement
III. Logic design
IV. HDL coding
(A) I-II-III-IV
(B) II-I-III-IV
(C) II-III-I-IV
(D) IV-I-II-III

Level II

1. Both top-down and bottom-up approaches are combined for a successful design because:
(A) they eliminate the need for transistor level design
(B) they provide better ideas to the designer to avoid some problems before starting a design
(C) they reduce the need for verification at each level
(D) they reduce the overall cost of manufacturing the IC

2. The Gajski chart represents the following domains of design flow.
(A) Logic, structural, and transistor domain
(B) Behavioural, structural, and geometrical layout domain
(C) Circuit, layout, and logic domain
(D) Functional, layout, and behavioural domain

3. VLSI design methodologies are:
(A) full custom design style and semi-custom design style
(B) logic level design style and transistor level design style
(C) gate array logic style and standard cell array
(D) transistor level design style and FPGA style

4. Standard cell based design belongs to:
(A) full custom design style.
(B) pre-designed library.
(C) semi-custom design style.
(D) designer's library.

5. What would be the highest level of representation among the following in VLSI design:
(A) functional design
(B) layout design
(C) logic design
(D) system specification

Answer Key

Level I

1. (B) **2.** (C) **3.** (C) **4.** (A) **5.** (B)

Level II

1. (B) **2.** (B) **3.** (A) **4.** (C) **5.** (D)